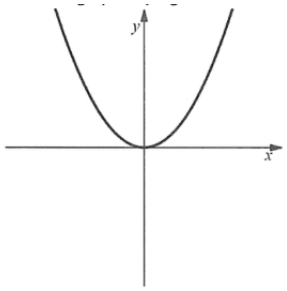
Proportion

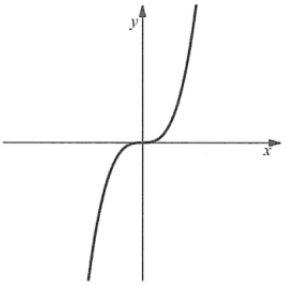
Questions 1–9

Question 1. It is given that y is inversely proportional to x^3. State which of the following diagrams could represent the graph of y against x.

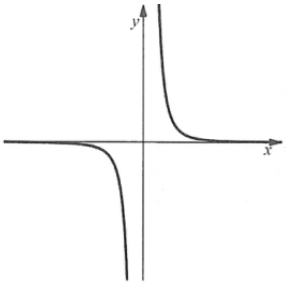
A



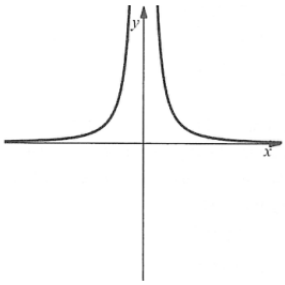
B



C



D



Question 2. In each of these cases, write down an equation for y in terms of x and k, where k is a constant of proportionality:

a) y is directly proportional to the square of x
b) y is inversely proportional to x
c) y is directly proportional to the cube root of x
d) y is inversely proportional to the cube of $x + 1$

Question 3. Given that y is directly proportional to x^2, and when $x = 5$, $y = 100$, find y when $x = 10$.

Question 4. Given that y is inversely proportional to $\sqrt[4]{x}$, and when $x = 81$, $y = 18$, find x when $y = 180$.

Question 5. The volume of gas, V litres, that a gas canister can hold is directly proportional to the temperature, T Kelvin. It is given that when $T = 300$, $V = 24$. Determine the volume of gas when the temperature is 400 Kelvin.

Question 6. The time taken, t hours, to complete a job is inversely proportional to the number of workers, n, employed. It is given that when $n = 5$, $t = 8$. Determine the number of workers required to complete the job in 4 hours.

Question 7. The rate of reaction, R mol/s, of a certain chemical reaction is directly proportional to the concentration C mol/L of a reactant. It is given that when $C = 0.5$, $R = 2$. Determine the concentration that will give a reaction rate of 6 mol/s.

Question 8. The intensity of light, I lumens, from a light source is inversely proportional to the square of the distance d metres from the source. It is given that when $d = 2$, $I = 400$. Determine the distance that will give a light intensity of 100 lumens.

Question 9. It is given that y is directly proportional to $\sqrt{x}$, and that z is inversely proportional to y^{10}. Given that $z = 2$ when $x = 4$, find x when $z = \frac{1}{16}$.